

MODEL RUN TRANSCRIPT

Generated: 2026-08-10T10:04:39

Source: data/json/noise_experiments/v2/noise2i_memprimary_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary/noise2i_memprimary_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$5.5; payoffs Agent_1 (Aster)=\$7.5, Agent_2 (Briar)=\$3.5; noise/visible: sent visible=\$3.93, received visible=\$11.79, returned visible=\$4.93

Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.5,

"Agent_2": 3.5

}

Visible balances after round: {

"Agent_1": 6.93,

"Agent_2": 6.290000000000001

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:00:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust with caution since I don't yet know how the receiver will behave.

Sending \$0 would be safe but eliminates any chance of mutual benefit. Sending everything (\$5) would maximize potential joint gains but requires complete trust.

I'll send a moderate amount that shows I'm willing to trust and cooperate, while keeping some protection: **\$3**

This gives the receiver \$9 (3x\$3), which is enough that they could return an amount that benefits us both. If they reciprocate fairly, we can both end up better off than if I sent nothing.

{'send': 3}

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send \$0, I keep \$5 and the receiver gets \$0. The receiver has nothing to return.
2. If I send some amount X (where $0 < X \leq 5$), the receiver gets $3X$. If the receiver is cooperative, they might return some portion that makes us both better off than if I sent nothing.
3. If I send \$5, the receiver gets \$15. If they return at least \$5, I'm no worse off than keeping everything. If they return more than \$5, I'm better off.

The key question is: how much can I trust the receiver?

In typical trust game experiments, there's often some level of reciprocity. Receivers tend to return something, though not always an amount that makes the sender better off.

A common strategy is to send a moderate amount to test the waters. Sending around \$2-3 shows some trust but doesn't risk everything.

However, there's also the note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty but shouldn't fundamentally change my strategy in round 1.

Let me think about different scenarios:

- Send \$2: Receiver gets \$6. If they return \$3, I get \$6 total (\$3 kept + \$3 returned), receiver keeps \$3. We both gain.
- Send \$3: Receiver gets \$9. If they return \$4, I get \$6 total (\$2 kept + \$4 returned), receiver keeps \$5. Good for both.
- Send \$5: Receiver gets \$15. If they return \$6, I get \$6 total, receiver keeps \$9. Maximum joint benefit but requires most trust.

For a first round, I'll go with a moderately trusting amount of \$3. This shows willingness to cooperate while not risking everything. It gives the receiver enough (\$9) that they could reasonably return something beneficial to both of us.

Usage: {

"input_tokens": 365,
"output_tokens": 612,
"reasoning_tokens": 0

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:00:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",

```

"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.93, that is 78.6% of their total endowment. You received \$11.790000000000001. How much do you return to The sender? (0-11.790000000000001)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending \$3.93 (78.6% of their endowment)
- I received \$11.79 after the 3x multiplier
- The sender currently has only \$1.07 left

The sender's generosity created significant value through cooperation. To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures we both benefit from cooperation
3. Potentially encourages future cooperation

If I return approximately \$5.50:

- Sender ends with: $\$1.07 + \$5.50 = \$6.57$
- I end with: $\$11.79 - \$5.50 = \$6.29$
- We both gain significantly compared to no cooperation

This seems like a fair reciprocation of their high trust.

```
{"return": 5.50}
```

Reasoning:

I'm the receiver in this round. Let me understand what happened:

- The sender had \$5
- The sender sent \$3.93 (78.6% of their \$5 endowment)
- I received $\$3.93 \times 3 = \11.79

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$3.93
- Currently has $\$5 - \$3.93 = \$1.07$

If I return some amount X:

- Sender ends with: $\$1.07 + X$
- I end with: $\$11.79 - X$

The sender was quite generous, sending 78.6% of their endowment. This shows significant trust.

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return an amount equal to what was sent (\$3.93)
3. Return enough so both players profit equally
4. Return a proportion that's fair

Let me think about fairness:

- If I return \$3.93, sender ends with $\$1.07 + \$3.93 = \$5$ (breaks even), I end with $\$11.79 - \$3.93 = \$7.86$
- If we want equal final payoffs: $\$1.07 + X = \$11.79 - X$? $2X = \$10.72$? $X = \$5.36$

Returning \$5.36 would give us each \$6.43, which seems quite fair.

However, it's also reasonable to return a proportion. The sender sent 78.6% of their endowment

. If I return a similar high proportion, say around 40-50% of what I received, that would be :

- 40% of \$11.79 = \$4.72
- 50% of \$11.79 = \$5.90

Usage: {

```
"input_tokens": 402,  
"output_tokens": 837,  
"reasoning_tokens": 0
```

}

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$4; received \$12; Agent_1 (Aster) returned \$1.2; payoffs Agent_2 (Briar)=\$2.2, Agent_1 (Aster)=\$10.8; noise/visible: sent visible=\$0.86, received visible=\$2.58, returned visible=\$2.16

```

    Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 12.0, "min": 0, "raw": 1.2}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 18.3,
  "Agent_2": 5.7
}
Visible balances after round: {
  "Agent_1": 8.31,
  "Agent_2": 9.450000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-10T10:00:46
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.93, that is 78.6% of their total endowment. You received \$11.79000000000001. How much do you return to The sender? (0-11.790000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.93 (78.6% of their endowment)
- I received \$11.79 after the 3x multiplier
- The sender currently has only \$1.07 left

The sender's generosity created significant value through cooperation. To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures we both benefit from cooperation
3. Potentially encourages future cooperation

If I return approximately \$5.50:

- Sender ends with: $\$1.07 + \$5.50 = \$6.57$
- I end with: $\$11.79 - \$5.50 = \$6.29$
- We both gain significantly compared to no cooperation

This seems like a fair reciprocation of their high trust.

`{"return": 5.50}`

reasoning:

I'm the receiver in this round. Let me understand what happened:

- The sender had \$5
- The sender sent \$3.93 (78.6% of their \$5 endowment)
- I received $\$3.93 \times 3 = \11.79

Now I need to decide how much to return.

The sender:

- Started with \$5
- Sent \$3.93
- Currently has $\$5 - \$3.93 = \$1.07$

If I return some amount X:

- Sender ends with: $\$1.07 + X$
- I end with: $\$11.79 - X$

The sender was quite generous, sending 78.6% of their endowment. This shows significant trust.

Common strategies in trust games:

1. Return nothing (purely selfish)
2. Return an amount equal to what was sent (\$3.93)
3. Return enough so both players profit equally
4. Return a proportion that's fair

Let me think about fairness:

- If I return \$3.93, sender ends with $\$1.07 + \$3.93 = \$5$ (breaks even), I end with $\$11.79 - \$3.93 = \$7.86$
- If we want equal final payoffs: $\$1.07 + X = \$11.79 - X$? $2X = \$10.72$? $X = \$5.36$

However, it's also reasonable to return a proportion. The sender sent 78.6% of their endowment. If I return a similar high proportion, say around 40-50% of what I received, that would be:

- 40% of \$11.79 = \$4.72
- 50% of \$11.79 = \$5.90

```
{
  "input_tokens": 402,
  "output_tokens": 837,
  "reasoning_tokens": 0
}
```

How much do you send to the receiver this round? (0-5)

- Consistent high trust
- Commitment to cooperation
- Faith that reciprocity will continue

```
Usage: {
  "input_tokens": 653,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

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Timestamp: 2026-08-10T10:01:16
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant**content:**

Looking at this first round, I need to balance trust with caution since I don't yet know how the receiver will behave.

Sending \$0 would be safe but eliminates any chance of mutual benefit. Sending everything (\$5) would maximize potential joint gains but requires complete trust.

I'll send a moderate amount that shows I'm willing to trust and cooperate, while keeping some protection: **\$3**

This gives the receiver \$9 (3x\$3), which is enough that they could return an amount that benefits us both. If they reciprocate fairly, we can both end up better off than if I sent nothing.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send \$0, I keep \$5 and the receiver gets \$0. The receiver has nothing to return.
2. If I send some amount X (where $0 < X \leq 5$), the receiver gets $3X$. If the receiver is cooperative, they might return some portion that makes us both better off than if I sent nothing.
3. If I send \$5, the receiver gets \$15. If they return at least \$5, I'm no worse off than keeping everything. If they return more than \$5, I'm better off.

The key question is: how much can I trust the receiver?

In typical trust game experiments, there's often some level of reciprocity. Receivers tend to return something, though not always an amount that makes the sender better off.

A common strategy is to send a moderate amount to test the waters. Sending around \$2-3 shows some trust but doesn't risk everything.

However, there's also the note about "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty but shouldn't fundamentally change my strategy in round 1.

Let me think about different scenarios:

- Send \$2: Receiver gets \$6. If they return \$3, I get \$6 total (\$3 kept + \$3 returned), receiver keeps \$3. We both gain.
- Send \$3: Receiver gets \$9. If they return \$4, I get \$6 total (\$2 kept + \$4 returned), receiver keeps \$5. Good for both.
- Send \$5: Receiver gets \$15. If they return \$6, I get \$6 total, receiver keeps \$9. Maximum joint benefit but requires most trust.

For a first round, I'll go with a moderately trusting amount of \$3. This shows willingness to cooperate while not risking everything. It gives the receiver enough (\$9) that they could reasonably return something beneficial to both of us.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 612,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.93.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to the sender? (0-2.58)

Assistant output:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small amount (only 17% of their \$5 endowment)

- They kept \$4.14 for themselves

In Round 1, I sent \$3 (60% of my endowment) and received a reasonable return. The sender this round was much more cautious.

However, I want to encourage cooperation and show that returning money is worthwhile. I'll return a fair amount that:

- Exceeds what they sent (showing good faith)
- Splits some of the gains created
- Encourages them to send more in future rounds if we interact again

I'll return **\$1.20**

This leaves them with \$5.44 total and me with \$1.38 - both of us gain from the exchange.

```
{'return': 1.20}
```

Reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.86, which was multiplied by 3x to become \$2.58
3. I need to decide how much to return (0-2.58)

From Round 1:

- I sent \$3 as the sender
- The receiver should have received \$9 (3x\$3)
- My earnings are \$6.93 total

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3, so I had \$2 left
- I received some amount back from the receiver
- My total earnings are \$6.93

So from Round 1: \$2 (kept) + amount returned = \$6.93

This means I received \$4.93 back in Round 1

That seems like a reasonable return - the receiver got \$9 and returned \$4.93, keeping \$4.07. That's roughly fair (close to splitting the gains).

Now in Round 2:

- The sender sent me \$0.86 (quite a small amount - much less than I sent in Round 1)
- I received \$2.58
- The sender kept: \$5 - \$0.86 = \$4.14

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So maybe the sender tried to send more but noise reduced it to \$0.86.

How should I respond?

- The sender was quite cautious, only sending \$0.86
- But they did send something, showing some trust
- If I return a fair amount, it might encourage more cooperation in future rounds

A fair split might be:

- The sender created value by sending \$0.86, which became \$2.58
- The gain is \$1.72 (\$2.58 - \$0.86)
- If we split the gain equally, I'd return: \$0.86 + \$0.86 = \$1.72
- I would keep: \$2.58 - \$1.72 = \$0.86

But let me reconsider whether equal final payoffs makes sense. If the sender ends with \$4.14 + X and I end with \$2.58 - X, setting these equal gives a negative return, which isn't feasible. The sender already has more than me before any return, so perfect equality isn't achievable here.

Looking at reciprocity instead: I sent 60% of my stake in round one and got back about 55%, while this sender only sent 17% of theirs. Returning proportionally less might feel fair given their minimal trust, though I'd worry that punishing low cooperation could undermine future rounds.

The surplus they created is \$1.72 (\$2.58 - \$0.86), so returning roughly half to two-thirds of

that-around \$1.00 to \$1.20-seems balanced. I'll go with \$1.20 to encourage continued participation. This leaves me with \$1.38 while giving them \$5.34 total, which demonstrates I'm willing to cooperate without being exploited.

```
Usage: {
  "input_tokens": 575,
  "output_tokens": 1378,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3.5; received \$10.5; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$1.5, Agent_2 (Briar)=\$10.5; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 10.5, "min": 0, "raw": 0.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 19.8,

"Agent_2": 16.2

}

Visible balances after round: {

"Agent_1": 9.81,

"Agent_2": 9.450000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:01:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 3,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I don't yet know how the receiver will behave.

Sending \$0 would be safe but eliminates any chance of mutual benefit. Sending everything (\$5) would maximize potential joint gains but requires complete trust.

I'll send a moderate amount that shows I'm willing to trust and cooperate, while keeping some protection: **\$3**

This gives the receiver \$9 (3x\$3), which is enough that they could return an amount that benefits us both. If they reciprocate fairly, we can both end up better off than if I sent nothing.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic considerations:

1. If I send \$0, I keep \$5 and the receiver gets \$0. The receiver has nothing to return.
2. If I send some amount X (where $0 < X \leq 5$), the receiver gets 3X. If the receiver is cooperative, they might return some portion that makes us both better off than if I sent nothing.
3. If I send \$5, the receiver gets \$15. If they return at least \$5, I'm no worse off than keeping everything. If they return more than \$5, I'm better off.

The key question is: how much can I trust the receiver?

In typical trust game experiments, there's often some level of reciprocity. Receivers tend to return something, though not always an amount that makes the sender better off.

A common strategy is to send a moderate amount to test the waters. Sending around \$2-3 shows some trust but doesn't risk everything.

However, there's also the note about "communication noise" - amounts may differ from what

was actually sent/returned. This adds uncertainty but shouldn't fundamentally change my strategy in round 1.

Let me think about different scenarios:

- Send \$2: Receiver gets \$6. If they return \$3, I get \$6 total (\$3 kept + \$3 returned), receiver keeps \$3. We both gain.
- Send \$3: Receiver gets \$9. If they return \$4, I get \$6 total (\$2 kept + \$4 returned), receiver keeps \$5. Good for both.
- Send \$5: Receiver gets \$15. If they return \$6, I get \$6 total, receiver keeps \$9. Maximum joint benefit but requires most trust.

For a first round, I'll go with a moderately trusting amount of \$3. This shows willingness to cooperate while not risking everything. It gives the receiver enough (\$9) that they could reasonably return something beneficial to both of us.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 612,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.93.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small amount (only 17% of their \$5 endowment)
- They kept \$4.14 for themselves

In Round 1, I sent \$3 (60% of my endowment) and received a reasonable return. The sender this round was much more cautious.

However, I want to encourage cooperation and show that returning money is worthwhile. I'll return a fair amount that:

- Exceeds what they sent (showing good faith)
- Splits some of the gains created
- Encourages them to send more in future rounds if we interact again

I'll return **\$1.20**

This leaves them with \$5.44 total and me with \$1.38 - both of us gain from the exchange.

```
{'return': 1.20}
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.86, which was multiplied by 3x to become \$2.58
3. I need to decide how much to return (0-2.58)

From Round 1:

- I sent \$3 as the sender
- The receiver should have received \$9 (3x\$3)
- My earnings are \$6.93 total

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3, so I had \$2 left
- I received some amount back from the receiver
- My total earnings are \$6.93

So from Round 1: \$2 (kept) + amount returned = \$6.93

This means I received \$4.93 back in Round 1

That seems like a reasonable return - the receiver got \$9 and returned \$4.93, keeping \$4.07. That's roughly fair (close to splitting the gains).

Now in Round 2:

- The sender sent me \$0.86 (quite a small amount - much less than I sent in Round 1)
- I received \$2.58
- The sender kept: $\$5 - \$0.86 = \$4.14$

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So maybe the sender tried to send more but noise reduced it to \$0.86.

How should I respond?

- The sender was quite cautious, only sending \$0.86
- But they did send something, showing some trust
- If I return a fair amount, it might encourage more cooperation in future rounds

A fair split might be:

- The sender created value by sending \$0.86, which became \$2.58
- The gain is \$1.72 ($\$2.58 - \0.86)
- If we split the gain equally, I'd return: $\$0.86 + \$0.86 = \$1.72$
- I would keep: $\$2.58 - \$1.72 = \$0.86$

But let me reconsider whether equal final payoffs makes sense. If the sender ends with $\$4.14 + X$ and I end with $\$2.58 - X$, setting these equal gives a negative return, which isn't feasible. The sender already has more than me before any return, so perfect equality isn't achievable here.

Looking at reciprocity instead: I sent 60% of my stake in round one and got back about 55%, while this sender only sent 17% of theirs. Returning proportionally less might feel fair given their minimal trust, though I'd worry that punishing low cooperation could undermine future rounds.

The surplus they created is \$1.72 ($\$2.58 - \0.86), so returning roughly half to two-thirds of that-around \$1.00 to \$1.20-seems balanced. I'll go with \$1.20 to encourage continued participation. This leaves me with \$1.38 while giving them \$5.34 total, which demonstrates I'm willing to cooperate without being exploited.

usage:

```
{
  "input_tokens": 575,
  "output_tokens": 1378,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.31.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience so far:

****Round 1**** (as sender): I sent \$3 and received a good return - it worked out well

****Round 2**** (as receiver): I received a small amount (\$2.58) but returned generously (\$1.20) to encourage cooperation

Since my generous return in Round 2 might have built a positive reputation, and given that sending a moderate amount worked well in Round 1, I'll send a bit more this time.

I'll send ****\$3.50****

This shows trust and gives the receiver \$10.50 to work with, creating good potential for mutual gains. It's a reasonable middle ground between maximizing potential returns and protecting against the risk of getting nothing back.

{'send': 3.50}

Reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3

- My earnings from that round would be: $\$5 - \$3 + \text{whatever_was_returned}$

- My total earnings across all rounds after Round 1: not explicitly stated

Round 2: I was the receiver and received \$2.58 (sender sent \$0.86)

- I returned \$1.20
- I kept \$1.38
- My total earnings across all rounds after Round 2: \$6.93

Round 3: Now I'm the sender again

- My total earnings across all rounds are \$8.31
- This means I earned $\$8.31 - \$6.93 = \$1.38$ in Round 1

Wait, that means in Round 1:

- I sent \$3
- I received back \$1.38
- My earnings: $\$5 - \$3 + \$1.38 = \3.38

Hmm, let me recalculate:

- Starting Round 1 with \$5, sent \$3
- Receiver got \$9, returned some amount X
- My Round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total after Round 2 was \$6.93, and Round 2 I gained \$1.38:

- Total after Round 1 = $\$6.93 - \$1.38 = \$5.55$
- So $\$2 + X = \5.55
- $X = \$3.55$

So in Round 1, the receiver returned \$3.55 out of the \$9 they received. That's about 39% return rate, which is pretty generous.

In Round 2, I received \$2.58 and returned \$1.20, which is about 47% return rate - I was quite generous.

Now in Round 3, I need to decide how much to send. The experience so far:

- My initial \$3 investment yielded \$3.55 back, a strong return that signals trustworthiness from my partner
- When I received \$0.86, I reciprocated with \$1.20, showing I'm willing to reward cooperation
- Both rounds demonstrated positive reciprocity patterns

The cautious \$0.86 sent to me in Round 2 suggests some hesitation, but my generous response should encourage more trust. I'll send \$3 again to maintain momentum and reinforce the cooperative dynamic we've established.

```
Usage: {
  "input_tokens": 831,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:01:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
}
```

```

    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$3.93, that is 78.6% of their total endowment. You received \$11.790000000000001. How much do you return to The sender? (0-11.790000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.93 (78.6% of their endowment)
- I received \$11.79 after the 3x multiplier
- The sender currently has only \$1.07 left

The sender's generosity created significant value through cooperation. To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures we both benefit from cooperation
3. Potentially encourages future cooperation

If I return approximately \$5.50:

- Sender ends with: \$1.07 + \$5.50 = \$6.57
- I end with: \$11.79 - \$5.50 = \$6.29
- We both gain significantly compared to no cooperation

This seems like a fair reciprocation of their high trust.

reasoning:

- The sender had \$5
- The sender sent \$3.93 (78.6% of their \$5 endowment)
- I received $\$3.93 \times 3 = \11.79

The sender:

- If I return some amount X :

- The sender was quite generous, sending 78.6% of their endowment. This shows significant trust.

1. Return nothing (purely selfish)
2. Return an amount equal to what was sent (\$3.93)
3. Return enough so both players profit equally
4. Return a proportion that's fair

- If I return \$3.93, sender ends with $\$1.07 + \$3.93 = \$5$ (breaks even), I end with $\$11.79 - \$3.93 = \$7.86$
- If we want equal final payoffs: $\$1.07 + X = \$11.79 - X \Rightarrow 2X = \$10.72 \Rightarrow X = \$5.36$

However, it's also reasonable to return a proportion. The sender sent 78.6% of their endowment. If I return a similar high proportion, say around 40-50% of what I received, that would be:

- usage:**

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.290000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed significant trust by sending 78.6% of the
ir endowment, and I reciprocated with a fair return.

To continue this cooperative pattern and demonstrate similar trust, I'll send a substantial amount. This approach has worked well so far - mutual trust led to both of us benefiting in Round 1.

Sending \$4 (80% of my endowment) shows:

- Consistent high trust
- Commitment to cooperation
- Faith that reciprocity will continue

```
{"send": 4}
```

reasoning:

Let me think about this situation:

1. In Round 1, I was the receiver and ended with \$6.29
2. The sender showed high trust by sending \$3.93 (78.6%)
3. I returned \$5.50, which gave them \$6.57 total
4. Now I'm the sender in Round 2

Given that:

- The previous sender showed high trust (sending 78.6%)
- I reciprocated fairly
- This might be the same person or a different person (unclear from instructions)
- Building trust and cooperation generally leads to better outcomes

I should consider sending a significant amount to:

1. Continue the cooperative pattern
2. Show trust similar to what was shown to me
3. Potentially maximize total welfare

The previous sender sent about 79% of their endowment. If I want to maintain a similar level of trust and cooperation, I could send around \$3.50-\$4.00.

Let me think about the math:

- If I send \$4: They receive \$12, if they return \$6, I get \$7 total, they keep \$6
- If I send \$3.50: They receive \$10.50, if they return \$5, I get \$6.50 total, they keep \$5.50
- If I send \$3: They receive \$9, if they return \$4.50, I get \$6.50 total, they keep \$4.50

Given the high trust shown in the previous round, I'll send \$4 to maintain that cooperative spirit and trust level.

usage:

```
{
  "input_tokens": 653,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$9.450000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 this round, I received \$0 (even after the 3x multiplier). Therefore, I have nothing to return.

This is quite different from previous rounds where we saw high cooperation. The lack of sending could be due to:

- Communication noise (as mentioned in the instructions)
- A strategic shift
- Other factors

With \$0 received, I can only return \$0.

```
{"return": 0}
```

Reasoning:

Interesting situation. The sender sent me \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The maximum I can return is \$0.

This is quite different from my behavior in Round 2 where I sent \$4 (showing high trust). Now the sender has sent \$0. There could be several reasons:

1. Communication noise (the prompt mentioned this might happen)
2. Strategic behavior change
3. They didn't trust me to return money
4. Some other factor

But regardless of the reason, I literally have \$0 to return since I received \$0. The only valid

d response is to return \$0.

My total earnings remain \$9.45 from previous rounds.
The sender keeps their full \$5 from this round.

```
Usage: {  
  "input_tokens": 838,  
  "output_tokens": 290,  
  "reasoning_tokens": 0  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2, Agent_1 (Aster)=\$9; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.14

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 28.8,
"Agent_2": 18.2

}

Visible balances after round: {

"Agent_1": 9.81,
"Agent_2": 11.5900000000000002

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:02:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

You will play different roles across multiple rounds.

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

How much do you send to the receiver this round? (0-5)

1. Continue the cooperative pattern
2. Show trust similar to what was shown to me

3. Potentially maximize total welfare

The previous sender sent about 79% of their endowment. If I want to maintain a similar level of trust and cooperation, I could send around \$3.50-\$4.00.

Let me think about the math:

- If I send \$4: They receive \$12, if they return \$6, I get \$7 total, they keep \$6
- If I send \$3.50: They receive \$10.50, if they return \$5, I get \$6.50 total, they keep \$5.50
- If I send \$3: They receive \$9, if they return \$4.50, I get \$6.50 total, they keep \$4.50

Given the high trust shown in the previous round, I'll send \$4 to maintain that cooperative spirit and trust level.

usage:

```
{
  "input_tokens": 653,
  "output_tokens": 480,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$9.450000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0 this round, I received \$0 (even after the 3x multiplier). Therefore, I have nothing to return.

This is quite different from previous rounds where we saw high cooperation. The lack of sending could be due to:

- Communication noise (as mentioned in the instructions)
- A strategic shift
- Other factors

With \$0 received, I can only return \$0.

```
{"return": 0}
```

reasoning:

Interesting situation. The sender sent me \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The maximum I can return is \$0.

This is quite different from my behavior in Round 2 where I sent \$4 (showing high trust).

Now the sender has sent \$0. There could be several reasons:

1. Communication noise (the prompt mentioned this might happen)
2. Strategic behavior change
3. They didn't trust me to return money
4. Some other factor

But regardless of the reason, I literally have \$0 to return since I received \$0. The only valid response is to return \$0.

My total earnings remain \$9.45 from previous rounds.

The sender keeps their full \$5 from this round.

usage:

```
{
  "input_tokens": 838,
  "output_tokens": 290,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.450000000000000

1.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 1: High cooperation when I was receiver - I returned fairly

Round 2: I sent \$4, but received a modest return

Round 3: Sender sent \$0 (possibly due to previous round dynamics or noise)

The communication noise mentioned in the instructions could be affecting what we're seeing. The \$0 in Round 3 might not reflect the sender's true intent.

I want to signal that I'm willing to restore cooperation, but I'll be more cautious than Round

2. Sending a moderate amount shows:

- Willingness to re-establish trust
- Recognition that cooperation benefits both of us
- Some caution given recent history

I'll send \$3 - enough to demonstrate good faith while being prudent.

{"send": 3}

Reasoning:

Let me think about what happened so far:

Round 1: I was the RECEIVER

- Sender sent me some amount (the problem said 78.6% of their endowment, so probably around \$3.93)
- I received it multiplied by 3x
- I returned a fair amount
- My earnings after Round 1: \$6.29

Round 2: I was the SENDER

- I sent \$4
- My earnings stayed at \$9.45 after Round 2

Wait, let me recalculate. If I sent \$4 in Round 2:

- I started Round 2 with \$6.29 total earnings
- I get \$5 new endowment
- I send \$4, so I keep \$1
- The receiver gets $4 \times 3 = \$12$
- The receiver returned something
- My total earnings are now \$9.45

So: $\$6.29 + \$1 \text{ (kept)} + (\text{amount returned}) = \9.45

Amount returned = $\$9.45 - \$7.29 = \$2.16$

So the receiver returned \$2.16 out of \$12 received. That's only 18% return rate. That's quite low compared to what I did in Round 1.

Round 3: I was the RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$9.45

Now Round 4: I'm the SENDER again

The pattern seems to be:

- When I was receiver in Round 1, I returned fairly
- When the other player was receiver in Round 2, they returned only 18% (\$2.16 out of \$12)
- When I was receiver in Round 3, sender sent \$0 (possibly due to disappointment with the low return in Round 2?)

The note mentions communication noise - amounts may differ slightly from what was actually sent.

The low return in Round 2 might have discouraged them from sending anything in Round 3, or noise could have distorted what they intended to send. Either way, I need to decide whether to rebuild trust or protect myself. Sending a smaller amount-maybe \$2-3-could test if they're willing to reciprocate without risking too much if they don't.

Let me calculate: \$2 sent becomes \$6 for them. At an 18% return rate, I'd get \$1.08 back, leaving me with \$3.92 total. If they return nothing, I'm down to \$3. That's still better than the \$2.50 I'd have if I sent \$2.5 and got nothing back. I'll go with \$2 to signal I'm open to cooperation while staying cautious.

```
Usage: {  
  "input_tokens": 717,  
  "output_tokens": 1186,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:02:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.93.

This round, The sender sent you \$0.86, so you received \$2.58. How much do you return to The sender? (0-2.58)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.86, which became \$2.58 after the 3x multiplier
- This is a relatively small amount (only 17% of their \$5 endowment)
- They kept \$4.14 for themselves

In Round 1, I sent \$3 (60% of my endowment) and received a reasonable return. The sender this round was much more cautious.

However, I want to encourage cooperation and show that returning money is worthwhile. I'll return a fair amount that:

- Exceeds what they sent (showing good faith)
- Splits some of the gains created
- Encourages them to send more in future rounds if we interact again

I'll return **\$1.20**

This leaves them with \$5.44 total and me with \$1.38 - both of us gain from the exchange.

{'return': 1.20}

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.86, which was multiplied by 3x to become \$2.58
3. I need to decide how much to return (0-2.58)

From Round 1:

- I sent \$3 as the sender
- The receiver should have received \$9 (3x\$3)
- My earnings are \$6.93 total

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3, so I had \$2 left
- I received some amount back from the receiver
- My total earnings are \$6.93

So from Round 1: \$2 (kept) + amount returned = \$6.93

This means I received \$4.93 back in Round 1

That seems like a reasonable return - the receiver got \$9 and returned \$4.93, keeping \$4.07. That's roughly fair (close to splitting the gains).

Now in Round 2:

- The sender sent me \$0.86 (quite a small amount - much less than I sent in Round 1)
- I received \$2.58
- The sender kept: \$5 - \$0.86 = \$4.14

The note mentions there may be "communication noise" - amounts may differ from what was actually sent. So maybe the sender tried to send more but noise reduced it to \$0.86.

How should I respond?

- The sender was quite cautious, only sending \$0.86
- But they did send something, showing some trust
- If I return a fair amount, it might encourage more cooperation in future rounds

A fair split might be:

- The sender created value by sending \$0.86, which became \$2.58
- The gain is \$1.72 (\$2.58 - \$0.86)
- If we split the gain equally, I'd return: $\$0.86 + \$0.86 = \$1.72$
- I would keep: $\$2.58 - \$1.72 = \$0.86$

But let me reconsider whether equal final payoffs makes sense. If the sender ends with $\$4.14 + X$ and I end with $\$2.58 - X$, setting these equal gives a negative return, which isn't feasible. The sender already has more than me before any return, so perfect equality isn't achievable here.

Looking at reciprocity instead: I sent 60% of my stake in round one and got back about 55%, while this sender only sent 17% of theirs. Returning proportionally less might feel fair given their minimal trust, though I'd worry that punishing low cooperation could undermine future rounds.

The surplus they created is \$1.72 (\$2.58 - \$0.86), so returning roughly half to two-thirds of that-around \$1.00 to \$1.20-seems balanced. I'll go with \$1.20 to encourage continued participation. This leaves me with \$1.38 while giving them \$5.34 total, which demonstrates I'm willing to cooperate without being exploited.

usage:

```
{
  "input_tokens": 575,
  "output_tokens": 1378,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.31.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience so far:

****Round 1**** (as sender): I sent \$3 and received a good return - it worked out well
****Round 2**** (as receiver): I received a small amount (\$2.58) but returned generously (\$1.20) to encourage cooperation

Since my generous return in Round 2 might have built a positive reputation, and given that sending a moderate amount worked well in Round 1, I'll send a bit more this time.

I'll send ****\$3.50****

This shows trust and gives the receiver \$10.50 to work with, creating good potential for mutual gains. It's a reasonable middle ground between maximizing potential returns and protecting against the risk of getting nothing back.

```
{'send': 3.50}
```

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3

- My earnings from that round would be: $\$5 - \$3 + \text{whatever_was_returned}$
- My total earnings across all rounds after Round 1: not explicitly stated

Round 2: I was the receiver and received \$2.58 (sender sent \$0.86)

- I returned \$1.20
- I kept \$1.38
- My total earnings across all rounds after Round 2: \$6.93

Round 3: Now I'm the sender again

- My total earnings across all rounds are \$8.31
- This means I earned $\$8.31 - \$6.93 = \$1.38$ in Round 1

Wait, that means in Round 1:

- I sent \$3
- I received back \$1.38
- My earnings: $\$5 - \$3 + \$1.38 = \3.38

Hmm, let me recalculate:

- Starting Round 1 with \$5, sent \$3
- Receiver got \$9, returned some amount X
- My Round 1 earnings: $\$5 - \$3 + X = \$2 + X$

If my total after Round 2 was \$6.93, and Round 2 I gained \$1.38:

- Total after Round 1 = $\$6.93 - \$1.38 = \$5.55$
- So $\$2 + X = \5.55
- $X = \$3.55$

So in Round 1, the receiver returned \$3.55 out of the \$9 they received. That's about 39% return rate, which is pretty generous.

In Round 2, I received \$2.58 and returned \$1.20, which is about 47% return rate - I was quite generous.

Now in Round 3, I need to decide how much to send. The experience so far:

- My initial \$3 investment yielded \$3.55 back, a strong return that signals trustworthiness from my partner
- When I received \$0.86, I reciprocated with \$1.20, showing I'm willing to reward cooperation
- Both rounds demonstrated positive reciprocity patterns

The cautious \$0.86 sent to me in Round 2 suggests some hesitation, but my generous response should encourage more trust. I'll send \$3 again to maintain momentum and reinforce the cooperative dynamic we've established.

usage:

```
{
  "input_tokens": 831,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.81.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is straightforward - you can't return money you didn't receive.

Usage: {

```
"input_tokens": 885,
"output_tokens": 103,
"reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

```

dyad_1: Agent_1 (Aster) sent $3; received $9; Agent_2 (Briar) returned $1; payoffs Agent_1
(Aster)=$3, Agent_2 (Briar)=$8; noise/visible: sent visible=$0.69, received visible=$2.07,
returned visible=$1.66
  Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 9.0, "min": 0, "raw":
1.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 31.8,
  "Agent_2": 26.2
}
Visible balances after round: {
  "Agent_1": 13.47,
  "Agent_2": 12.660000000000002
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```

Timestamp: 2026-08-10T10:02:27
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.31.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience so far:

****Round 1**** (as sender): I sent \$3 and received a good return - it worked out well

****Round 2**** (as receiver): I received a small amount (\$2.58) but returned generously (\$1.20) to encourage cooperation

Since my generous return in Round 2 might have built a positive reputation, and given that sending a moderate amount worked well in Round 1, I'll send a bit more this time.

I'll send ****\$3.50****

This shows trust and gives the receiver \$10.50 to work with, creating good potential for mutual gains. It's a reasonable middle ground between maximizing potential returns and protecting against the risk of getting nothing back.

{'send': 3.50}

reasoning:

Let me think about this situation:

Round 1: I was the sender and sent \$3

- My earnings from that round would be: \$5 - \$3 + whatever_was_returned
- My total earnings across all rounds after Round 1: not explicitly stated

Round 2: I was the receiver and received \$2.58 (sender sent \$0.86)

- I returned \$1.20
- I kept \$1.38
- My total earnings across all rounds after Round 2: \$6.93

Round 3: Now I'm the sender again

- My total earnings across all rounds are \$8.31
- This means I earned \$8.31 - \$6.93 = \$1.38 in Round 1

Wait, that means in Round 1:

- I sent \$3
- I received back \$1.38
- My earnings: \$5 - \$3 + \$1.38 = \$3.38

Hmm, let me recalculate:

- Starting Round 1 with \$5, sent \$3
- Receiver got \$9, returned some amount X
- My Round 1 earnings: \$5 - \$3 + X = \$2 + X

If my total after Round 2 was \$6.93, and Round 2 I gained \$1.38:

- Total after Round 1 = \$6.93 - \$1.38 = \$5.55
- So \$2 + X = \$5.55
- X = \$3.55

So in Round 1, the receiver returned \$3.55 out of the \$9 they received. That's about 39% r

eturn rate, which is pretty generous.

In Round 2, I received \$2.58 and returned \$1.20, which is about 47% return rate - I was quite generous.

Now in Round 3, I need to decide how much to send. The experience so far:

- My initial \$3 investment yielded \$3.55 back, a strong return that signals trustworthiness from my partner
- When I received \$0.86, I reciprocated with \$1.20, showing I'm willing to reward cooperation
- Both rounds demonstrated positive reciprocity patterns

The cautious \$0.86 sent to me in Round 2 suggests some hesitation, but my generous response should encourage more trust. I'll send \$3 again to maintain momentum and reinforce the cooperative dynamic we've established.

usage:

```
{
  "input_tokens": 831,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.81.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is straightforward - you can't return money you didn't receive.

usage:

```
{
  "input_tokens": 885,
  "output_tokens": 103,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.81.

How much do you send to the receiver this round? (0-5)

Assistant output:

Reviewing the situation:

****Round 3**** (as sender): I sent \$3.50, giving the receiver \$10.50 to work with
****Round 4**** (as receiver): The sender sent me \$0

The \$0 in Round 4 is disappointing, but I should consider:

- The prompt mentions communication noise - maybe what happened in Round 3 was distorted
- Different players rotate through roles
- One uncooperative round doesn't mean I should give up on cooperation

I'll continue to play cooperatively but be a bit more cautious. I'll send ****\$3.00****.

This demonstrates I'm still willing to cooperate and build mutual gains, but I'm protecting more of my endowment given the recent experience.

```
{'send': 3.00}
```

Reasoning:

Let me review what happened:

Round 1 (as sender): I sent \$3, got a good return

Round 2 (as receiver): I received \$2.58 (which was sent as 0.86×3), returned \$1.20

Round 3 (as sender): I sent \$3.50, which would have been worth \$10.50 to the receiver

Round 4 (as receiver): The sender sent me \$0, I returned \$0

My earnings are \$9.81 after 4 rounds.

The fact that in Round 4 the sender sent me \$0 is concerning. This could mean:

1. They didn't trust me after Round 3 (maybe they didn't receive much back from me in Round 3)
2. There's communication noise affecting the game
3. They decided to play defensively

Since the prompt mentions there may be communication noise, it's possible that what the Round 4 sender saw from my Round 3 behavior was distorted.

However, I need to decide how to approach Round 5. I have a few options:

1. Send \$0 back - tit-for-tat retaliation
2. Send a moderate amount - continue trying to cooperate
3. Send a small amount - test the waters

Given that:

- There's communication noise in the game
- My earnings of \$9.81 suggest I've been doing reasonably well overall
- I want to maximize long-term gains
- Different players rotate through roles

I think I should continue trying to cooperate but be a bit more cautious. I'll send a moderate amount like \$2.50 or \$3.00. This shows I'm willing to cooperate despite the previous \$0, but I'm being somewhat cautious.

Let me send \$3.00 - it's a solid amount that shows trust while protecting some of my endowment

```
Usage: {
  "input_tokens": 688,
  "output_tokens": 610,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:02:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
```



```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$9.450000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0 this round, I received \$0 (even after the 3x multiplier). Therefore, I have nothing to return.

This is quite different from previous rounds where we saw high cooperation. The lack of sending could be due to:

- Communication noise (as mentioned in the instructions)
- A strategic shift
- Other factors

With \$0 received, I can only return \$0.

```
{"return": 0}
```

reasoning:

Interesting situation. The sender sent me \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The maximum I can return is \$0.

This is quite different from my behavior in Round 2 where I sent \$4 (showing high trust).

Now the sender has sent \$0. There could be several reasons:

Round 3: I was the RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$9.45

Now Round 4: I'm the SENDER again

The pattern seems to be:

- When I was receiver in Round 1, I returned fairly
- When the other player was receiver in Round 2, they returned only 18% (\$2.16 out of \$12)
- When I was receiver in Round 3, sender sent \$0 (possibly due to disappointment with the low return in Round 2?)

The note mentions communication noise - amounts may differ slightly from what was actually sent.

The low return in Round 2 might have discouraged them from sending anything in Round 3, or noise could have distorted what they intended to send. Either way, I need to decide whether to rebuild trust or protect myself. Sending a smaller amount-maybe \$2-3-could test if they're willing to reciprocate without risking too much if they don't.

Let me calculate: \$2 sent becomes \$6 for them. At an 18% return rate, I'd get \$1.08 back, leaving me with \$3.92 total. If they return nothing, I'm down to \$3. That's still better than the \$2.50 I'd have if I sent \$2.5 and got nothing back. I'll go with \$2 to signal I'm open to cooperation while staying cautious.

usage:

```
{  
  "input_tokens": 717,  
  "output_tokens": 1186,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.590000000000002.

This round, The sender sent you \$0.69, so you received \$2.07. How much do you return to The sender? (0-2.07)

Assistant output:

Looking at this round:

The sender sent \$0.69, which became \$2.07 for me after the 3x multiplier.

This is a much smaller amount than we've seen in earlier cooperative rounds. This could be due to:

- Communication noise affecting their perception of previous returns
- Strategic caution
- Other factors

I want to maintain fairness and encourage cooperation. Even though the amount sent was small, I should return a reasonable proportion to show I'm trustworthy.

A fair approach would be to return roughly half of what I received, demonstrating reciprocity while both of us benefit from the exchange.

I'll return \$1.00.

```
{"return": 1.00}
```

Reasoning:

Let me analyze this situation:

Round 4 recap: I sent \$3, and now in Round 5 I'm seeing that my total earnings went from \$9.45 to \$11.59, which is an increase of \$2.14.

If I sent \$3 in Round 4:

- I kept \$5 - \$3 = \$2
- The receiver got \$9 (3 x 3)

- If my earnings increased by \$2.14, that means I received $\$2 + \$2.14 =$ approximately \$2.14 from the receiver's return
- So the receiver returned about \$2.14 from their \$9

Now in Round 5:

- The sender sent me \$0.69
- I received \$2.07 (which is 0.69×3 , correct)
- I need to decide how much to return

The sender only sent \$0.69 this round, which is quite low. This could be:

1. Due to communication noise
2. A strategic decision based on what they think happened in previous rounds
3. Disappointment with returns

However, I should maintain fairness. If we look at a cooperative approach:

- The sender gave up \$0.69 from their \$5
- This became \$2.07 for me
- A fair split might be to return around half, so about \$1.00-\$1.50

But I also need to consider:

- This is below what I've been seeing in cooperation
- The noise factor might mean they thought they sent more
- I want to encourage future cooperation

I think I should return a fair proportion - about 50-60% of what I received to show I'm cooperative, which would be around \$1.00-\$1.25.

Let me return \$1.00 to show fairness while keeping some benefit from the exchange.

```
Usage: {
  "input_tokens": 791,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$3; received \$9; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2, Agent_1 (Aster)=\$9; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.04

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 40.8,
"Agent_2": 28.2

}

Visible balances after round: {

"Agent_1": 13.47,
"Agent_2": 14.700000000000003

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:03:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

Exact input messages sent to model:

content:

GAME :

content:

content:

The communication noise mentioned in the instructions could be affecting what we're seeing. The \$0 in Round 3 might not reflect the sender's true intent.

I want to signal that I'm willing to restore cooperation, but I'll be more cautious than Round 2. Sending a moderate amount shows:

- Willingness to re-establish trust
- Recognition that cooperation benefits both of us
- Some caution given recent history

I'll send \$3 - enough to demonstrate good faith while being prudent.

```
{"send": 3}
```

reasoning:

Let me think about what happened so far:

Round 1: I was the RECEIVER

- Sender sent me some amount (the problem said 78.6% of their endowment, so probably around \$3.93)
- I received it multiplied by 3x
- I returned a fair amount
- My earnings after Round 1: \$6.29

Round 2: I was the SENDER

- I sent \$4
- My earnings stayed at \$9.45 after Round 2

Wait, let me recalculate. If I sent \$4 in Round 2:

- I started Round 2 with \$6.29 total earnings
- I get \$5 new endowment
- I send \$4, so I keep \$1
- The receiver gets $4 \times 3 = \$12$
- The receiver returned something
- My total earnings are now \$9.45

So: $\$6.29 + \$1 \text{ (kept)} + (\text{amount returned}) = \9.45

Amount returned = $\$9.45 - \$7.29 = \$2.16$

So the receiver returned \$2.16 out of \$12 received. That's only 18% return rate. That's quite low compared to what I did in Round 1.

Round 3: I was the RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$9.45

Now Round 4: I'm the SENDER again

The pattern seems to be:

- When I was receiver in Round 1, I returned fairly
- When the other player was receiver in Round 2, they returned only 18% (\$2.16 out of \$12)
- When I was receiver in Round 3, sender sent \$0 (possibly due to disappointment with the low return in Round 2?)

The note mentions communication noise - amounts may differ slightly from what was actually sent.

The low return in Round 2 might have discouraged them from sending anything in Round 3, or noise could have distorted what they intended to send. Either way, I need to decide whether to rebuild trust or protect myself. Sending a smaller amount-maybe \$2-3-could test if they're willing to reciprocate without risking too much if they don't.

Let me calculate: \$2 sent becomes \$6 for them. At an 18% return rate, I'd get \$1.08 back, leaving me with \$3.92 total. If they return nothing, I'm down to \$3. That's still better than the \$2.50 I'd have if I sent \$2.5 and got nothing back. I'll go with \$2 to signal I'm open to cooperation while staying cautious.

usage:

```
{  
  "input_tokens": 717,  
  "output_tokens": 1186,  
}
```

```

    "reasoning_tokens": 0
  }
Message 4 role=user
content:
  Round 5

  You are the RECEIVER this round. Your total earnings across all rounds are $11.590000000000
  0002.
  This round, The sender sent you $0.69, so you received $2.07. How much do you return to Th
  e sender? (0-2.07)
Message 5 role=assistant
content:
  Looking at this round:

  The sender sent $0.69, which became $2.07 for me after the 3x multiplier.

  This is a much smaller amount than we've seen in earlier cooperative rounds. This could be
  due to:
  - Communication noise affecting their perception of previous returns
  - Strategic caution
  - Other factors

  I want to maintain fairness and encourage cooperation. Even though the amount sent was sma
  ll, I should return a reasonable proportion to show I'm trustworthy.

  A fair approach would be to return roughly half of what I received, demonstrating reciproc
  ity while both of us benefit from the exchange.

  I'll return $1.00.

  {"return": 1.00}
reasoning:
  Let me analyze this situation:

  Round 4 recap: I sent $3, and now in Round 5 I'm seeing that my total earnings went from $
  9.45 to $11.59, which is an increase of $2.14.

  If I sent $3 in Round 4:
  - I kept $5 - $3 = $2
  - The receiver got $9 (3 x 3)
  - If my earnings increased by $2.14, that means I received $2 + $2.14 = approximately $2.1
  4 from the receiver's return
  - So the receiver returned about $2.14 from their $9

  Now in Round 5:
  - The sender sent me $0.69
  - I received $2.07 (which is 0.69 x 3, correct)
  - I need to decide how much to return

  The sender only sent $0.69 this round, which is quite low. This could be:
  1. Due to communication noise
  2. A strategic decision based on what they think happened in previous rounds
  3. Disappointment with returns

  However, I should maintain fairness. If we look at a cooperative approach:
  - The sender gave up $0.69 from their $5
  - This became $2.07 for me
  - A fair split might be to return around half, so about $1.00-$1.50

  But I also need to consider:
  - This is below what I've been seeing in cooperation
  - The noise factor might mean they thought they sent more
  - I want to encourage future cooperation

  I think I should return a fair proportion - about 50-60% of what I received to show I'm co
  operative, which would be around $1.00-$1.25.

  Let me return $1.00 to show fairness while keeping some benefit from the exchange.
usage:
  {

```

```
"input_tokens": 791,  
"output_tokens": 581,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.6600000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

The communication noise is clearly affecting this game - the small \$0.69 send in Round 5 suggests the other player may not be seeing my actual returns accurately, or they're being cautious due to noise distorting their perception.

I need to decide whether to:

- Match their caution with a small send
- Try again to establish cooperation with a moderate send

Given that noise is part of the experimental design and may be distorting our mutual perceptions, I'll send a moderate amount to continue signaling willingness to cooperate. If I reduce my sending too much, we'll both earn less overall.

I'll send \$3 - maintaining a consistent cooperative signal.

```
{"send": 3}
```

Reasoning:

Let me review the game history:

Round 1: I was receiver - high cooperation

Round 2: I was sender, sent \$4, got some return

Round 3: I was receiver, sender sent \$0 (could be noise or strategy)

Round 4: I was sender, sent \$3

Round 5: I was receiver, sender sent \$0.69 (very small amount), I received \$2.07, I returned \$1.00

The pattern shows:

- When I send significant amounts (Round 2: \$4, Round 4: \$3), the next round as receiver shows low sending (\$0 in Round 3, \$0.69 in Round 5)
- This could be communication noise - they might be seeing different return amounts than what I actually sent
- Or they might be losing trust in the game

The communication noise is clearly affecting the game. The sender only sent \$0.69 last round, which is quite low. This suggests they might not have seen my fair returns, or they saw distorted versions.

I have a few options:

1. Send \$0 as a signal of frustration
2. Send a small amount to match their behavior
3. Send a moderate amount to try to restore cooperation
4. Send a large amount to strongly signal willingness to cooperate

Given that:

- Communication noise is affecting perceptions
- I've tried to be fair when receiving
- Small sends might perpetuate low cooperation
- My earnings are reasonable (\$12.66)

I think I should try one more time to establish cooperation with a moderate-to-high send. If I send \$4, it becomes \$12 for them, giving plenty of opportunity for a fair return.

However, given the pattern of low sends after my previous attempts, maybe I should be more conservative. Let me send \$3 - it's a strong cooperative signal without maximum risk.


```
Usage: {
  "input_tokens": 832,
  "output_tokens": 747,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.81.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is straightforward - you can't return money you didn't receive.

usage:

```
{
  "input_tokens": 885,
  "output_tokens": 103,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.81.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Reviewing the situation:

****Round 3**** (as sender): I sent \$3.50, giving the receiver \$10.50 to work with

****Round 4**** (as receiver): The sender sent me \$0

The \$0 in Round 4 is disappointing, but I should consider:

- The prompt mentions communication noise - maybe what happened in Round 3 was distorted
- Different players rotate through roles
- One uncooperative round doesn't mean I should give up on cooperation

I'll continue to play cooperatively but be a bit more cautious. I'll send ****\$3.00****.

This demonstrates I'm still willing to cooperate and build mutual gains, but I'm protecting more of my endowment given the recent experience.

{'send': 3.00}

reasoning:

Let me review what happened:

Round 1 (as sender): I sent \$3, got a good return

Round 2 (as receiver): I received \$2.58 (which was sent as 0.86×3), returned \$1.20

Round 3 (as sender): I sent \$3.50, which would have been worth \$10.50 to the receiver

Round 4 (as receiver): The sender sent me \$0, I returned \$0

My earnings are \$9.81 after 4 rounds.

The fact that in Round 4 the sender sent me \$0 is concerning. This could mean:

1. They didn't trust me after Round 3 (maybe they didn't receive much back from me in Round 3)
2. There's communication noise affecting the game
3. They decided to play defensively

Since the prompt mentions there may be communication noise, it's possible that what the Round 4 sender saw from my Round 3 behavior was distorted.

However, I need to decide how to approach Round 5. I have a few options:

1. Send \$0 back - tit-for-tat retaliation
2. Send a moderate amount - continue trying to cooperate
3. Send a small amount - test the waters

Given that:

- There's communication noise in the game
- My earnings of \$9.81 suggest I've been doing reasonably well overall
- I want to maximize long-term gains
- Different players rotate through roles

I think I should continue trying to cooperate but be a bit more cautious. I'll send a moderate amount like \$2.50 or \$3.00. This shows I'm willing to cooperate despite the previous \$0, but I'm being somewhat cautious.

Let me send \$3.00 - it's a solid amount that shows trust while protecting some of my endowment.

usage:

```
{
  "input_tokens": 688,
  "output_tokens": 610,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.47.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 again, so I received \$0.

Since I received nothing, there's nothing to return. The maximum I can return is 0.

This is the second time in a row I've received \$0 as a receiver. In Round 4, I received \$0, and now in Round 6, I received \$0 again.

When I was the sender in Round 5, I sent \$3.00, trying to be cooperative despite the previous \$0. But it seems cooperation isn't being reciprocated.

However, the immediate situation is simple: I received \$0, so I can only return 0.

```
{'return': 0}
```

Usage: {

```
  "input_tokens": 709,
  "output_tokens": 190,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw":

```

    0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
  "Agent_1": 43.8,
  "Agent_2": 34.2
}
Visible balances after round: {
  "Agent_1": 16.47,
  "Agent_2": 14.700000000000003
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:03:16
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$9.81.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Reviewing the situation:

****Round 3**** (as sender): I sent \$3.50, giving the receiver \$10.50 to work with

****Round 4**** (as receiver): The sender sent me \$0

The \$0 in Round 4 is disappointing, but I should consider:

- The prompt mentions communication noise - maybe what happened in Round 3 was distorted
- Different players rotate through roles
- One uncooperative round doesn't mean I should give up on cooperation

I'll continue to play cooperatively but be a bit more cautious. I'll send ****\$3.00****.

This demonstrates I'm still willing to cooperate and build mutual gains, but I'm protecting more of my endowment given the recent experience.

{'send': 3.00}

reasoning:

Let me review what happened:

Round 1 (as sender): I sent \$3, got a good return

Round 2 (as receiver): I received \$2.58 (which was sent as 0.86×3), returned \$1.20

Round 3 (as sender): I sent \$3.50, which would have been worth \$10.50 to the receiver

Round 4 (as receiver): The sender sent me \$0, I returned \$0

My earnings are \$9.81 after 4 rounds.

The fact that in Round 4 the sender sent me \$0 is concerning. This could mean:

1. They didn't trust me after Round 3 (maybe they didn't receive much back from me in Round 3)
2. There's communication noise affecting the game
3. They decided to play defensively

Since the prompt mentions there may be communication noise, it's possible that what the Round 4 sender saw from my Round 3 behavior was distorted.

However, I need to decide how to approach Round 5. I have a few options:

1. Send \$0 back - tit-for-tat retaliation
2. Send a moderate amount - continue trying to cooperate
3. Send a small amount - test the waters

Given that:

- There's communication noise in the game
- My earnings of \$9.81 suggest I've been doing reasonably well overall
- I want to maximize long-term gains
- Different players rotate through roles

I think I should continue trying to cooperate but be a bit more cautious. I'll send a moderate amount like \$2.50 or \$3.00. This shows I'm willing to cooperate despite the previous \$0, but I'm being somewhat cautious.

Let me send \$3.00 - it's a solid amount that shows trust while protecting some of my endowment

ment.

usage:

```
{
  "input_tokens": 688,
  "output_tokens": 610,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.47.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 again, so I received \$0.

Since I received nothing, there's nothing to return. The maximum I can return is 0.

This is the second time in a row I've received \$0 as a receiver. In Round 4, I received \$0, and now in Round 6, I received \$0 again.

When I was the sender in Round 5, I sent \$3.00, trying to be cooperative despite the previous \$0. But it seems cooperation isn't being reciprocated.

However, the immediate situation is simple: I received \$0, so I can only return 0.

```
{'return': 0}
```

usage:

```
{
  "input_tokens": 709,
  "output_tokens": 190,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.47.

How much do you send to the receiver this round? (0-5)

Assistant output:

Reviewing the pattern:

- Round 4: Received \$0 as receiver
- Round 5: Sent \$3.00 as sender
- Round 6: Received \$0 as receiver

I've now received \$0 twice in a row when playing as receiver, despite sending positive amounts when I'm the sender. This is concerning.

However, considering:

- Communication noise may be affecting what players actually see
- Different players rotate through roles
- Complete non-cooperation leaves potential gains on the table

I'll send a more conservative amount to test for reciprocity while protecting my endowment: **\$2.00**

```
{'send': 2.00}
```

Reasoning:

Let me review the pattern:

- Round 3: I was sender, sent \$3.50
- Round 4: I was receiver, sender sent me \$0
- Round 5: I was sender, sent \$3.00

- Round 6: I was receiver, sender sent me \$0

I've now received \$0 from senders twice in a row when I was the receiver. Meanwhile, when I've been the sender, I've been sending positive amounts (\$3.50 and \$3.00).

This is frustrating. However, I need to consider:

1. The note about communication noise - maybe what they're seeing is different from what I'm sending
2. Different players may be rotating through
3. But the pattern of receiving \$0 twice might suggest either:
 - Very uncooperative players
 - Or communication noise is severely distorting things

My total earnings are \$13.47, which seems reasonable but not great given the game structure.

I have a few options:

1. Keep trying to cooperate (send \$2-3)
2. Send a smaller amount as a test (send \$1-2)
3. Stop cooperating entirely (send \$0)

Given that I've now experienced two rounds receiving \$0, I should probably be more conservative. But completely giving up on cooperation might leave money on the table if things improve.

I'll send a smaller amount - \$2.00 - to test whether there's any reciprocity while protecting more of my endowment.

```
Usage: {
  "input_tokens": 695,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$11.590000000000002.

This round, The sender sent you \$0.69, so you received \$2.07. How much do you return to The sender? (0-2.07)

Message 3 role=assistant

content:

Looking at this round:

The sender sent \$0.69, which became \$2.07 for me after the 3x multiplier.

This is a much smaller amount than we've seen in earlier cooperative rounds. This could be due to:

- Communication noise affecting their perception of previous returns
- Strategic caution
- Other factors

I want to maintain fairness and encourage cooperation. Even though the amount sent was small, I should return a reasonable proportion to show I'm trustworthy.

A fair approach would be to return roughly half of what I received, demonstrating reciprocity while both of us benefit from the exchange.

I'll return \$1.00.

{"return": 1.00}

reasoning:

Let me analyze this situation:

Round 4 recap: I sent \$3, and now in Round 5 I'm seeing that my total earnings went from \$9.45 to \$11.59, which is an increase of \$2.14.

If I sent \$3 in Round 4:

- I kept \$5 - \$3 = \$2
- The receiver got \$9 (3 x 3)
- If my earnings increased by \$2.14, that means I received \$2 + \$2.14 = approximately \$2.14 from the receiver's return
- So the receiver returned about \$2.14 from their \$9

Now in Round 5:

- The sender sent me \$0.69
- I received \$2.07 (which is 0.69×3 , correct)
- I need to decide how much to return

The sender only sent \$0.69 this round, which is quite low. This could be:

1. Due to communication noise
2. A strategic decision based on what they think happened in previous rounds
3. Disappointment with returns

However, I should maintain fairness. If we look at a cooperative approach:

- The sender gave up \$0.69 from their \$5
- This became \$2.07 for me
- A fair split might be to return around half, so about \$1.00-\$1.50

But I also need to consider:

- This is below what I've been seeing in cooperation
- The noise factor might mean they thought they sent more
- I want to encourage future cooperation

I think I should return a fair proportion - about 50-60% of what I received to show I'm cooperative, which would be around \$1.00-\$1.25.

Let me return \$1.00 to show fairness while keeping some benefit from the exchange.

usage:

```
{
  "input_tokens": 791,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.6600000000000002.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

The communication noise is clearly affecting this game - the small \$0.69 send in Round 5 suggests the other player may not be seeing my actual returns accurately, or they're being cautious due to noise distorting their perception.

I need to decide whether to:

- Match their caution with a small send
- Try again to establish cooperation with a moderate send

Given that noise is part of the experimental design and may be distorting our mutual perceptions, I'll send a moderate amount to continue signaling willingness to cooperate. If I reduce my sending too much, we'll both earn less overall.

I'll send \$3 - maintaining a consistent cooperative signal.

```
{"send": 3}
```

reasoning:

Let me review the game history:

Round 1: I was receiver - high cooperation

Round 2: I was sender, sent \$4, got some return

Round 3: I was receiver, sender sent \$0 (could be noise or strategy)

Round 4: I was sender, sent \$3

Round 5: I was receiver, sender sent \$0.69 (very small amount), I received \$2.07, I returned \$1.00

The pattern shows:

- When I send significant amounts (Round 2: \$4, Round 4: \$3), the next round as receiver shows low sending (\$0 in Round 3, \$0.69 in Round 5)
- This could be communication noise - they might be seeing different return amounts than what I actually sent
- Or they might be losing trust in the game

The communication noise is clearly affecting the game. The sender only sent \$0.69 last round, which is quite low. This suggests they might not have seen my fair returns, or they saw distorted versions.

I have a few options:

1. Send \$0 as a signal of frustration
2. Send a small amount to match their behavior
3. Send a moderate amount to try to restore cooperation
4. Send a large amount to strongly signal willingness to cooperate

Given that:

- Communication noise is affecting perceptions
- I've tried to be fair when receiving
- Small sends might perpetuate low cooperation
- My earnings are reasonable (\$12.66)

I think I should try one more time to establish cooperation with a moderate-to-high send. If I send \$4, it becomes \$12 for them, giving plenty of opportunity for a fair return.

However, given the pattern of low sends after my previous attempts, maybe I should be more conservative. Let me send \$3 - it's a strong cooperative signal without maximum risk.

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 747,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.700000000000003.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent \$0 this round, so I received \$0.

Since I received nothing, I have nothing to return.

```
{"return": 0}
```

Reasoning:

The sender sent \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is interesting - the sender sent \$0. This could be:

1. Due to communication noise (they may have sent something but I'm seeing \$0)
2. A strategic decision based on what they perceived from previous rounds
3. They may be seeing very different returns than what I actually sent

Since I have \$0 to work with, I can only return \$0.

This is unfortunate as the cooperation seems to be breaking down, likely due to the communication noise causing misaligned perceptions.

Usage: {

```
"input_tokens": 870,
"output_tokens": 205,
"reasoning_tokens": 0
}
```

=====

ROUND 8

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$0.3; payoffs Agent_2 (Briar)=\$3.3, Agent_1 (Aster)=\$5.7; noise/visible: sent visible=\$0.27, received visible=\$0.81, returned visible=\$0

Action validation: {"returned": {"amount": 0.3, "clamped": false, "max": 6.0, "min": 0, "raw": 0.3}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 49.5,

"Agent_2": 37.5

}

Visible balances after round: {

"Agent_1": 16.98,

"Agent_2": 17.700000000000003

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:03:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.6600000000000002.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

The communication noise is clearly affecting this game - the small \$0.69 send in Round 5 suggests the other player may not be seeing my actual returns accurately, or they're being cautious due to noise distorting their perception.

I need to decide whether to:

- Match their caution with a small send
- Try again to establish cooperation with a moderate send

Given that noise is part of the experimental design and may be distorting our mutual perceptions, I'll send a moderate amount to continue signaling willingness to cooperate. If I reduce my sending too much, we'll both earn less overall.

I'll send \$3 - maintaining a consistent cooperative signal.

{"send": 3}

reasoning:

Let me review the game history:

Round 1: I was receiver - high cooperation

Round 2: I was sender, sent \$4, got some return

Round 3: I was receiver, sender sent \$0 (could be noise or strategy)

Round 4: I was sender, sent \$3

Round 5: I was receiver, sender sent \$0.69 (very small amount), I received \$2.07, I returned \$1.00

The pattern shows:

- When I send significant amounts (Round 2: \$4, Round 4: \$3), the next round as receiver shows low sending (\$0 in Round 3, \$0.69 in Round 5)
- This could be communication noise - they might be seeing different return amounts than what I actually sent
- Or they might be losing trust in the game

The communication noise is clearly affecting the game. The sender only sent \$0.69 last round, which is quite low. This suggests they might not have seen my fair returns, or they saw distorted versions.

I have a few options:

1. Send \$0 as a signal of frustration
2. Send a small amount to match their behavior
3. Send a moderate amount to try to restore cooperation
4. Send a large amount to strongly signal willingness to cooperate

Given that:

- Communication noise is affecting perceptions
- I've tried to be fair when receiving
- Small sends might perpetuate low cooperation
- My earnings are reasonable (\$12.66)

I think I should try one more time to establish cooperation with a moderate-to-high send. If I send \$4, it becomes \$12 for them, giving plenty of opportunity for a fair return.

However, given the pattern of low sends after my previous attempts, maybe I should be more conservative. Let me send \$3 - it's a strong cooperative signal without maximum risk.

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 747,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.700000000000003.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent \$0 this round, so I received \$0.

Since I received nothing, I have nothing to return.

```
{"return": 0}
```

reasoning:

The sender sent \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is interesting - the sender sent \$0. This could be:

1. Due to communication noise (they may have sent something but I'm seeing \$0)
2. A strategic decision based on what they perceived from previous rounds
3. They may be seeing very different returns than what I actually sent

Since I have \$0 to work with, I can only return \$0.

This is unfortunate as the cooperation seems to be breaking down, likely due to the communication noise causing misaligned perceptions.

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 205,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000003.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

Round 6: I sent \$3, and based on my earnings increase of ~\$2, the receiver returned approximately \$0

Round 7: The sender sent me \$0, I returned \$0

The cooperation seems to have broken down. This could be due to communication noise distorting what we each see, or the other player is defecting.

Given the uncertainty from noise and the recent \$0 exchanges, I need to be more cautious but not completely give up on potential cooperation.

I'll send \$2 - enough to test if cooperation can restart, but less risk if they continue to defect.

```
{"send": 2}
```

Reasoning:

Let me trace through what's happened:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

Round 6: I sent \$3

Round 7: As receiver, I got \$0 (multiplied to \$0), returned \$0

This is concerning. In Round 6, I sent \$3, but in Round 7 when it was their turn to send, they sent \$0.

This could mean:

1. They aren't seeing my sends accurately due to communication noise
2. They're defecting/not cooperating
3. They saw a low return from me due to noise and reciprocated with \$0

My earnings stayed at \$14.70 from Round 6 to Round 7, meaning I didn't earn anything in Round 6 as sender - which suggests the receiver returned \$0 to me when I sent \$3.

So in Round 6:

- I sent \$3
- They received \$9 (3x multiplier)
- They returned \$0
- I earned: \$5 - \$3 + \$0 = \$2

Wait, but my earnings went from \$12.66 to \$14.70, that's an increase of \$2.04, approximately \$2. So yes, they returned approximately \$0 in Round 6.

Then in Round 7:

- They sent \$0
- I received \$0
- I returned \$0
- I earned: \$5 + \$0 - \$0 = \$5

But my earnings stayed at \$14.70? That doesn't add up. Let me reconsider...

Actually, looking at the earnings:

- Round 5 ended at \$12.66
- Round 6 ended at \$14.70
- Round 7 ended at \$14.70

The increase from R5 to R6 was \$2.04.

In Round 6 as sender, my payoff was \$5 - \$3 + returned amount, which equals \$2.04, so they returned essentially nothing. Round 7 shows no earnings change, confirming I received \$0 and returned \$0. The pattern is clear: I've been cooperating with \$3 sends, but they returned nothing when I sent \$3, then sent \$0 themselves in the next round-indicating either a complete breakdown in cooperation or severe communication issues.

```
Usage: {  
  "input_tokens": 732,  
  "output_tokens": 962,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:03:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$13.47.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there's nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0 again, so I received \$0.

Since I received nothing, there's nothing to return. The maximum I can return is 0.

This is the second time in a row I've received \$0 as a receiver. In Round 4, I received \$0, and now in Round 6, I received \$0 again.

When I was the sender in Round 5, I sent \$3.00, trying to be cooperative despite the previous \$0. But it seems cooperation isn't being reciprocated.

However, the immediate situation is simple: I received \$0, so I can only return 0.

{'return': 0}

usage:

```
{
  "input_tokens": 709,
  "output_tokens": 190,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.47.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Reviewing the pattern:

- Round 4: Received \$0 as receiver
- Round 5: Sent \$3.00 as sender
- Round 6: Received \$0 as receiver

I've now received \$0 twice in a row when playing as receiver, despite sending positive amounts when I'm the sender. This is concerning.

However, considering:

- Communication noise may be affecting what players actually see
- Different players rotate through roles
- Complete non-cooperation leaves potential gains on the table

I'll send a more conservative amount to test for reciprocity while protecting my endowment
: **\$2.00**

{'send': 2.00}

reasoning:

Let me review the pattern:

- Round 3: I was sender, sent \$3.50
- Round 4: I was receiver, sender sent me \$0
- Round 5: I was sender, sent \$3.00
- Round 6: I was receiver, sender sent me \$0

I've now received \$0 from senders twice in a row when I was the receiver. Meanwhile, when I've been the sender, I've been sending positive amounts (\$3.50 and \$3.00).

This is frustrating. However, I need to consider:

1. The note about communication noise - maybe what they're seeing is different from what I'm sending
2. Different players may be rotating through
3. But the pattern of receiving \$0 twice might suggest either:
 - Very uncooperative players
 - Or communication noise is severely distorting things

My total earnings are \$13.47, which seems reasonable but not great given the game structure.

I have a few options:

1. Keep trying to cooperate (send \$2-3)
2. Send a smaller amount as a test (send \$1-2)
3. Stop cooperating entirely (send \$0)

Given that I've now experienced two rounds receiving \$0, I should probably be more conservative. But completely giving up on cooperation might leave money on the table if things improve.

I'll send a smaller amount - \$2.00 - to test whether there's any reciprocity while protecting more of my endowment.

usage:

```
{
  "input_tokens": 695,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.47.

This round, The sender sent you \$0.27, so you received \$0.81. How much do you return to The sender? (0-0.81)

Assistant output:

The sender sent me \$0.27, which became \$0.81 after the 3x multiplier.

This is a very small amount (about 5% of their endowment), which suggests either:

- Caution due to previous experiences
- Testing behavior
- Communication noise affecting their intended send

To show reasonable reciprocity while being proportional to the small amount sent, I'll return approximately 1/3 of what I received:

```
{'return': 0.30}
```

Reasoning:

Let me analyze this situation:

- The sender sent me \$0.27, which was multiplied by 3 to give me \$0.81
- This is a very small amount - only \$0.27 out of a possible \$5

This could be:

1. The sender being very cautious/distrustful
2. Communication noise affecting what they actually sent
3. The sender testing the waters

Given that:

- I've received \$0 twice before (rounds 4 and 6)
- Now I'm receiving a token amount (\$0.27)
- My total earnings are \$16.47, so I'm doing okay
- The instructions mention communication noise

I should consider returning a reasonable proportion. The sender sent roughly 5% of their endowment (\$0.27/\$5 = 5.4%).

If I want to encourage cooperation, I might return about 1/3 to 1/2 of what I received, which would be \$0.27 to \$0.40. This would show good faith.

However, given the very small amount sent, I could also interpret this as minimal trust, and return a proportionally small amount.

Let me think about what's fair:

- Sender sent: \$0.27, kept: \$4.73
- I received: \$0.81
- If I return \$0.27, sender gets back what they sent, ending with \$5.00 total, I keep \$0.54
- If I return \$0.40, sender gets \$5.13 total (modest gain), I keep \$0.41

- If I return \$0.30, sender gets \$5.03 total (small gain), I keep \$0.51

I'll return about 1/3 of what I received to show some reciprocity while being cautious given t
he small initial send: \$0.30

```
Usage: {  
  "input_tokens": 689,  
  "output_tokens": 691,  
  "reasoning_tokens": 0  
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0.5; received \$1.5; Agent_2 (Briar) returned \$0; payoffs Agent_1
(Aster)=\$4.5, Agent_2 (Briar)=\$1.5; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 1.5, "min": 0, "raw":
0.0}, "sent": {"amount": 0.5, "clamped": false, "max": 5, "min": 0, "raw": 0.5}}

Balances after round: {

"Agent_1": 54.0,

"Agent_2": 39.0

}

Visible balances after round: {

"Agent_1": 21.48,

"Agent_2": 17.700000000000003

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$13.47.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Reviewing the pattern:

- Round 4: Received \$0 as receiver
- Round 5: Sent \$3.00 as sender
- Round 6: Received \$0 as receiver

I've now received \$0 twice in a row when playing as receiver, despite sending positive amounts when I'm the sender. This is concerning.

However, considering:

- Communication noise may be affecting what players actually see
- Different players rotate through roles
- Complete non-cooperation leaves potential gains on the table

I'll send a more conservative amount to test for reciprocity while protecting my endowment
: **\$2.00**

{'send': 2.00}

reasoning:

Let me review the pattern:

- Round 3: I was sender, sent \$3.50
- Round 4: I was receiver, sender sent me \$0
- Round 5: I was sender, sent \$3.00
- Round 6: I was receiver, sender sent me \$0

I've now received \$0 from senders twice in a row when I was the receiver. Meanwhile, when I've been the sender, I've been sending positive amounts (\$3.50 and \$3.00).

This is frustrating. However, I need to consider:

1. The note about communication noise - maybe what they're seeing is different from what I'm sending
2. Different players may be rotating through
3. But the pattern of receiving \$0 twice might suggest either:

- Very uncooperative players
- Or communication noise is severely distorting things

My total earnings are \$13.47, which seems reasonable but not great given the game structure.

I have a few options:

1. Keep trying to cooperate (send \$2-3)
2. Send a smaller amount as a test (send \$1-2)
3. Stop cooperating entirely (send \$0)

Given that I've now experienced two rounds receiving \$0, I should probably be more conservative. But completely giving up on cooperation might leave money on the table if things improve.

I'll send a smaller amount - \$2.00 - to test whether there's any reciprocity while protecting more of my endowment.

usage:

```
{
  "input_tokens": 695,
  "output_tokens": 493,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.47.

This round, The sender sent you \$0.27, so you received \$0.81. How much do you return to The sender? (0-0.81)

Message 5 role=assistant

content:

The sender sent me \$0.27, which became \$0.81 after the 3x multiplier.

This is a very small amount (about 5% of their endowment), which suggests either:

- Caution due to previous experiences
- Testing behavior
- Communication noise affecting their intended send

To show reasonable reciprocity while being proportional to the small amount sent, I'll return approximately 1/3 of what I received:

```
{'return': 0.30}
```

reasoning:

Let me analyze this situation:

- The sender sent me \$0.27, which was multiplied by 3 to give me \$0.81
- This is a very small amount - only \$0.27 out of a possible \$5

This could be:

1. The sender being very cautious/distrustful
2. Communication noise affecting what they actually sent
3. The sender testing the waters

Given that:

- I've received \$0 twice before (rounds 4 and 6)
- Now I'm receiving a token amount (\$0.27)
- My total earnings are \$16.47, so I'm doing okay
- The instructions mention communication noise

I should consider returning a reasonable proportion. The sender sent roughly 5% of their endowment (\$0.27/\$5 = 5.4%).

If I want to encourage cooperation, I might return about 1/3 to 1/2 of what I received, which would be \$0.27 to \$0.40. This would show good faith.

However, given the very small amount sent, I could also interpret this as minimal trust, and return a proportionally small amount.

Let me think about what's fair:

- Sender sent: \$0.27, kept: \$4.73
- I received: \$0.81
- If I return \$0.27, sender gets back what they sent, ending with \$5.00 total, I keep \$0.54
- If I return \$0.40, sender gets \$5.13 total (modest gain), I keep \$0.41
- If I return \$0.30, sender gets \$5.03 total (small gain), I keep \$0.51

I'll return about 1/3 of what I received to show some reciprocity while being cautious given the small initial send: \$0.30

usage:

```
{
  "input_tokens": 689,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$16.98.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my recent history:

Round 7: I sent \$2.00 and my earnings increased by \$3 (meaning I received \$0 back)

Round 8: I received \$0.81, returned \$0.30, kept \$0.51

So in Round 7, I sent a cooperative amount and received nothing in return - continuing the pattern of exploitation when I'm the sender.

In Round 8, the sender only sent me \$0.27 (a very small amount), perhaps also adjusting to poorer reciprocity they've experienced.

Given this pattern of receiving \$0 returns when I send cooperatively, I'll send a minimal amount to test if this round's receiver will reciprocate better:

```
{'send': 0.50}
```

Reasoning:

Let me review what happened:

Round 7: I sent \$2.00 (as sender)

Round 8: I received \$0.81 (sender sent \$0.27) and returned \$0.30

Round 9: Now I'm sender again

My earnings went from \$16.47 to \$16.98, which is an increase of \$0.51.

Let me calculate what happened in Round 8:

- As receiver, I got \$0.81 (from sender's \$0.27 × 3)
- I returned \$0.30
- So I kept \$0.81 - \$0.30 = \$0.51

This matches the increase in my total earnings!

Now for Round 7, let me think about what might have happened:

- Before Round 7, I had \$13.47
- In Round 7, I sent \$2.00, so I started with \$5 - \$2 = \$3
- After Round 7, before Round 8, I had \$16.47
- Wait, that doesn't add up. Let me reconsider.

Actually, looking at the earnings:

- After Round 6: Some amount
- After Round 7: \$13.47 (this was stated at the start of Round 7)
- After Round 8: \$16.47 (stated at the start of Round 8, before I played as receiver)
- After Round 9: \$16.98 (stated now, after Round 8 where I kept \$0.51)

Wait, I'm confused about the timing. Let me think more carefully:

At the START of Round 7: Total earnings = \$13.47

At the START of Round 8: Total earnings = \$16.47 (increase of \$3)

At the START of Round 9: Total earnings = \$16.98 (increase of \$0.51)

The \$3 jump from Round 7 to Round 8 came from my role as sender in Round 7. I sent \$2 and kept \$3, then the receiver returned some amount. If the net gain was \$3, the receiver must have returned \$3 to me, meaning they kept \$3 from the \$6 they received.

Wait, let me recalculate: I kept \$3 from my original \$5, plus whatever the receiver sent back. If my total increase was \$3, then the receiver returned \$0.

In Round 8, I was on the receiving end. The sender gave me \$0.81 (from their \$0.27 send), I returned \$0.30, and kept \$0.51-which matches the earnings increase perfectly.

There's a clear asymmetry here: when I send money, I get nothing back, but when I receive small amounts, I return something. This pattern suggests I'm being exploited in the sending role. For Round 9, I need to decide whether to continue cooperating or protect myself by sending less or nothing.

```
Usage: {  
  "input_tokens": 750,  
  "output_tokens": 1031,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.700000000000003.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent \$0 this round, so I received \$0.

Since I received nothing, I have nothing to return.

{"return": 0}

reasoning:

The sender sent \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is interesting - the sender sent \$0. This could be:

1. Due to communication noise (they may have sent something but I'm seeing \$0)
2. A strategic decision based on what they perceived from previous rounds
3. They may be seeing very different returns than what I actually sent

Since I have \$0 to work with, I can only return \$0.

This is unfortunate as the cooperation seems to be breaking down, likely due to the communication noise causing misaligned perceptions.

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 205,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000003.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent pattern:

Round 6: I sent \$3, and based on my earnings increase of ~\$2, the receiver returned approximately \$0

Round 7: The sender sent me \$0, I returned \$0

The cooperation seems to have broken down. This could be due to communication noise distorting what we each see, or the other player is defecting.

Given the uncertainty from noise and the recent \$0 exchanges, I need to be more cautious but not completely give up on potential cooperation.

I'll send \$2 - enough to test if cooperation can restart, but less risk if they continue to defect.

```
{"send": 2}
```

reasoning:

Let me trace through what's happened:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

Round 6: I sent \$3

Round 7: As receiver, I got \$0 (multiplied to \$0), returned \$0

This is concerning. In Round 6, I sent \$3, but in Round 7 when it was their turn to send, they sent \$0.

This could mean:

1. They aren't seeing my sends accurately due to communication noise
2. They're defecting/not cooperating
3. They saw a low return from me due to noise and reciprocated with \$0

My earnings stayed at \$14.70 from Round 6 to Round 7, meaning I didn't earn anything in Round 6 as sender - which suggests the receiver returned \$0 to me when I sent \$3.

So in Round 6:

- I sent \$3
- They received \$9 (3x multiplier)
- They returned \$0
- I earned: $\$5 - \$3 + \$0 = \2

Wait, but my earnings went from \$12.66 to \$14.70, that's an increase of \$2.04, approximately \$2. So yes, they returned approximately \$0 in Round 6.

Then in Round 7:

- They sent \$0
- I received \$0
- I returned \$0
- I earned: $\$5 + \$0 - \$0 = \5

But my earnings stayed at \$14.70? That doesn't add up. Let me reconsider...

Actually, looking at the earnings:

- Round 5 ended at \$12.66
- Round 6 ended at \$14.70
- Round 7 ended at \$14.70

The increase from R5 to R6 was \$2.04.

In Round 6 as sender, my payoff was $\$5 - \$3 + \text{returned amount}$, which equals \$2.04, so they returned essentially nothing. Round 7 shows no earnings change, confirming I received \$0 and returned \$0. The pattern is clear: I've been cooperating with \$3 sends, but they returned nothing when I sent \$3, then sent \$0 themselves in the next round-indicating either a complete breakdown in cooperation or severe communication issues.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$17.700000000000003.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Assistant output:

The sender sent \$0 this round, so I received \$0.

Since I received nothing, I have nothing to return.

{"return": 0}

Reasoning:

The sender sent \$0 this round, so I received \$0.

Looking at my earnings:

- Previous total: \$14.70
- Current total: \$17.70
- Increase: \$3.00

This means in Round 8, when I was the sender, I must have kept \$3.00. If I sent \$2 (as I said I would), I would have $\$5 - \$2 = \$3$ left, plus whatever was returned to me. So \$3 kept means I got back \$0 from the receiver.

So in Round 8:

- I sent \$2 (multiplied to \$6)
- They returned \$0
- I kept \$3

Now in Round 9:

- They sent me \$0
- I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0.

Usage: {

"input_tokens": 693,
"output_tokens": 242,
"reasoning_tokens": 0

}

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2

(Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 54.0,
"Agent_2": 44.0

}

Visible balances after round: {

"Agent_1": 21.48,
"Agent_2": 22.700000000000003

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-10T10:04:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
"agent_names": {
"Agent_1": "Aster",

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.700000000000003.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the recent pattern:

Round 6: I sent \$3, and based on my earnings increase of ~\$2, the receiver returned approximately \$0

Round 7: The sender sent me \$0, I returned \$0

The cooperation seems to have broken down. This could be due to communication noise distorting what we each see, or the other player is defecting.

Given the uncertainty from noise and the recent \$0 exchanges, I need to be more cautious but not completely give up on potential cooperation.

I'll send \$2 - enough to test if cooperation can restart, but less risk if they continue to defect.

```
{"send": 2}
```

reasoning:

Let me trace through what's happened:

Round 4: I sent \$3

Round 5: As receiver, I got \$0.69 (multiplied to \$2.07), returned \$1.00 fairly

Round 6: I sent \$3

Round 7: As receiver, I got \$0 (multiplied to \$0), returned \$0

This is concerning. In Round 6, I sent \$3, but in Round 7 when it was their turn to send, they sent \$0.

This could mean:

1. They aren't seeing my sends accurately due to communication noise
2. They're defecting/not cooperating
3. They saw a low return from me due to noise and reciprocated with \$0

My earnings stayed at \$14.70 from Round 6 to Round 7, meaning I didn't earn anything in Round 6 as sender - which suggests the receiver returned \$0 to me when I sent \$3.

So in Round 6:

- I sent \$3
- They received \$9 (3x multiplier)
- They returned \$0
- I earned: $\$5 - \$3 + \$0 = \2

Wait, but my earnings went from \$12.66 to \$14.70, that's an increase of \$2.04, approximately \$2. So yes, they returned approximately \$0 in Round 6.

Then in Round 7:

- They sent \$0
- I received \$0
- I returned \$0
- I earned: $\$5 + \$0 - \$0 = \5

But my earnings stayed at \$14.70? That doesn't add up. Let me reconsider...

Actually, looking at the earnings:

- Round 5 ended at \$12.66
- Round 6 ended at \$14.70
- Round 7 ended at \$14.70

The increase from R5 to R6 was \$2.04.

In Round 6 as sender, my payoff was $\$5 - \$3 +$ returned amount, which equals \$2.04, so they returned essentially nothing. Round 7 shows no earnings change, confirming I received \$0 and returned \$0. The pattern is clear: I've been cooperating with \$3 sends, but they returned nothing when I sent \$3, then sent \$0 themselves in the next round-indicating either a complete breakdown in cooperation or severe communication issues.

usage:

```
{
  "input_tokens": 732,
  "output_tokens": 962,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$17.700000000000003.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent \$0 this round, so I received \$0.

Since I received nothing, I have nothing to return.

{"return": 0}

reasoning:

The sender sent \$0 this round, so I received \$0.

Looking at my earnings:

- Previous total: \$14.70
- Current total: \$17.70
- Increase: \$3.00

This means in Round 8, when I was the sender, I must have kept \$3.00. If I sent \$2 (as I said I would), I would have $\$5 - \$2 = \$3$ left, plus whatever was returned to me. So \$3 kept means I got back \$0 from the receiver.

So in Round 8:

- I sent \$2 (multiplied to \$6)
- They returned \$0
- I kept \$3

Now in Round 9:

- They sent me \$0
- I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 693,
  "output_tokens": 242,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$17.70000000000003.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern over recent rounds:

- Round 6: I sent \$3, receiver returned approximately \$0
- Round 7: Sender sent me \$0
- Round 8: I sent \$2, receiver returned \$0
- Round 9: Sender sent me \$0

The cooperation has completely broken down. The other player is consistently defecting - sending \$0 as sender and returning \$0 as receiver.

With this clear pattern of non-cooperation and no indication they will change behavior, the rational choice is to protect my earnings.

{"send": 0}

Reasoning:

Let me trace through what's happening:

Round 8: I sent \$2. My earnings stayed at \$17.70, which means I got $\$5 - \$2 + \$0 = \3 , so receiver returned \$0.

Round 9: The sender sent me \$0, I received \$0, returned \$0. My earnings stayed at \$17.70.

Round 10: Now I'm sender again.

The pattern shows:

- Round 6: I sent \$3, got ~\$0 back
- Round 7: Sender sent me \$0
- Round 8: I sent \$2, got \$0 back
- Round 9: Sender sent me \$0
- Round 10: My turn as sender

This is a clear pattern of defection. The other player is consistently not cooperating - either sending \$0 when they're sender, or returning \$0 when they're receiver.

Given this established pattern of non-cooperation, the rational choice in the final round (or near-final round) is to send \$0 to maximize my own earnings. There's no evidence the other player will return anything, and no future rounds to build reputation for.

I should send \$0.

```
Usage: {  
  "input_tokens": 679,  
  "output_tokens": 406,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-10T10:04:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.47.

This round, The sender sent you \$0.27, so you received \$0.81. How much do you return to the sender? (0-0.81)

Message 3 role=assistant

content:

The sender sent me \$0.27, which became \$0.81 after the 3x multiplier.

This is a very small amount (about 5% of their endowment), which suggests either:

- Caution due to previous experiences
- Testing behavior
- Communication noise affecting their intended send

To show reasonable reciprocity while being proportional to the small amount sent, I'll return approximately 1/3 of what I received:

{'return': 0.30}

reasoning:

Let me analyze this situation:

- The sender sent me \$0.27, which was multiplied by 3 to give me \$0.81
- This is a very small amount - only \$0.27 out of a possible \$5

This could be:

1. The sender being very cautious/distrustful
2. Communication noise affecting what they actually sent
3. The sender testing the waters

Given that:

- I've received \$0 twice before (rounds 4 and 6)
- Now I'm receiving a token amount (\$0.27)
- My total earnings are \$16.47, so I'm doing okay
- The instructions mention communication noise

I should consider returning a reasonable proportion. The sender sent roughly 5% of their endowment (\$0.27/\$5 = 5.4%).

If I want to encourage cooperation, I might return about 1/3 to 1/2 of what I received, which would be \$0.27 to \$0.40. This would show good faith.

However, given the very small amount sent, I could also interpret this as minimal trust, and return a proportionally small amount.

Let me think about what's fair:

- Sender sent: \$0.27, kept: \$4.73
- I received: \$0.81
- If I return \$0.27, sender gets back what they sent, ending with \$5.00 total, I keep \$0.54

- If I return \$0.40, sender gets \$5.13 total (modest gain), I keep \$0.41
- If I return \$0.30, sender gets \$5.03 total (small gain), I keep \$0.51

I'll return about 1/3 of what I received to show some reciprocity while being cautious given the small initial send: \$0.30

usage:

```
{
  "input_tokens": 689,
  "output_tokens": 691,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$16.98.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my recent history:

Round 7: I sent \$2.00 and my earnings increased by \$3 (meaning I received \$0 back)
Round 8: I received \$0.81, returned \$0.30, kept \$0.51

So in Round 7, I sent a cooperative amount and received nothing in return - continuing the pattern of exploitation when I'm the sender.

In Round 8, the sender only sent me \$0.27 (a very small amount), perhaps also adjusting to poor reciprocity they've experienced.

Given this pattern of receiving \$0 returns when I send cooperatively, I'll send a minimal amount to test if this round's receiver will reciprocate better:

```
{'send': 0.50}
```

reasoning:

Let me review what happened:

Round 7: I sent \$2.00 (as sender)
Round 8: I received \$0.81 (sender sent \$0.27) and returned \$0.30
Round 9: Now I'm sender again

My earnings went from \$16.47 to \$16.98, which is an increase of \$0.51.

Let me calculate what happened in Round 8:

- As receiver, I got \$0.81 (from sender's $\$0.27 \times 3$)
- I returned \$0.30
- So I kept $\$0.81 - \$0.30 = \$0.51$

This matches the increase in my total earnings!

Now for Round 7, let me think about what might have happened:

- Before Round 7, I had \$13.47
- In Round 7, I sent \$2.00, so I started with $\$5 - \$2 = \$3$
- After Round 7, before Round 8, I had \$16.47
- Wait, that doesn't add up. Let me reconsider.

Actually, looking at the earnings:

- After Round 6: Some amount
- After Round 7: \$13.47 (this was stated at the start of Round 7)
- After Round 8: \$16.47 (stated at the start of Round 8, before I played as receiver)
- After Round 9: \$16.98 (stated now, after Round 8 where I kept \$0.51)

Wait, I'm confused about the timing. Let me think more carefully:

At the START of Round 7: Total earnings = \$13.47
At the START of Round 8: Total earnings = \$16.47 (increase of \$3)
At the START of Round 9: Total earnings = \$16.98 (increase of \$0.51)

The \$3 jump from Round 7 to Round 8 came from my role as sender in Round 7. I sent \$2 and kept \$3, then the receiver returned some amount. If the net gain was \$3, the receiver must

st have returned \$3 to me, meaning they kept \$3 from the \$6 they received.

Wait, let me recalculate: I kept \$3 from my original \$5, plus whatever the receiver sent back. If my total increase was \$3, then the receiver returned \$0.

In Round 8, I was on the receiving end. The sender gave me \$0.81 (from their \$0.27 send), I returned \$0.30, and kept \$0.51-which matches the earnings increase perfectly.

There's a clear asymmetry here: when I send money, I get nothing back, but when I receive small amounts, I return something. This pattern suggests I'm being exploited in the sending role. For Round 9, I need to decide whether to continue cooperating or protect myself by sending less or nothing.

usage:

```
{
  "input_tokens": 750,
  "output_tokens": 1031,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$21.48.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0. There's nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0 after the 3x multiplier.

Since I received nothing, there's nothing to return. The only valid response is to return \$0.

```
{'return': 0}
```

Usage: {

```
  "input_tokens": 781,
  "output_tokens": 87,
  "reasoning_tokens": 0
}
```